DAY 2 – SECURE CONNECTIVITY, WEB DELIVERY & TROUBLESHOOTING

Focus: HTTP/HTTPS, TLS certificates and handshake, SSH authentication, custom domains, and production incident response.

PART E – HTTP, HTTPS & TLS Analysis

Run and capture the outputs of: curl -I http://example.com, curl -I https://example.com, and curl -v https://example.com. (**Use a public test domain and do not include credentials in command output).**

21. In the response headers, identify the HTTP status line and at least four headers. Explain what each selected header tells you.

The HTTP status line contains the HTTP version, status code, and reason phrase (where applicable). It indicates the result of the server's response. The following are important HTTP response headers:

Content-Type: Specifies the media type of the response, such as text/html.

Cache-Control: Specifies caching instructions for browsers and intermediate caches.

Date: Indicates the date and time when the response was generated.

Server: May identify the web server software used by the server.

Location: Specifies the destination URL when the server redirects a request.

The actual status line and headers should be identified from the output of the curl commands.

22. The HTTP request redirects to HTTPS. Identify the redirect status code and the Location header, if present. Explain what the browser does next.

When an HTTP request is redirected to HTTPS, the server returns a redirection status code such as 301 (Moved Permanently), 302 (Found), 307 (Temporary Redirect), or 308 (Permanent Redirect).
The Location response header specifies the new HTTPS URL.
The browser reads the Location header and sends a new request to that HTTPS URL. It then establishes a secure TLS connection before requesting the website content.

23. Explain how TLS protects data in transit. Include server identity verification, encryption, and integrity in your answer.

Transport Layer Security (TLS) protects data transmitted between a client and a server. It provides three major security benefits:

Server Identity Verification: The server presents a digital certificate. The client verifies the certificate's trust chain, validity, and hostname to confirm the server's identity.

Encryption: TLS encrypts application data so that unauthorized parties cannot easily read it while it travels across the network.

Integrity: TLS uses cryptographic authentication mechanisms to detect unauthorized modification of transmitted data.
Together, these features help provide secure communication between browsers and websites.

24. Arrange and explain the major stages of a simplified TLS connection: Client Hello, Server Hello, certificate presentation and validation, key agreement, and encrypted application traffic. Mention that modern TLS versions can differ in handshake details.

The simplified TLS connection process includes the following stages:

- **Client Hello:** The client sends the supported TLS versions, cipher suites, and key agreement information to the server.
- **Server Hello:** The server selects the TLS version and cipher suite and provides the required key agreement information.
- **Certificate Presentation and Validation:** The server presents its digital certificate. The client validates the certificate chain, hostname, and validity period.
- **Key Agreement:** The client and server establish shared key material and derive session encryption keys.
- **Encrypted Application Traffic:** After the handshake is completed, application data such as HTTP requests and responses is transmitted securely using encryption and integrity protection.
  Note: This is a simplified explanation. The exact handshake sequence and details differ between modern TLS versions, such as TLS 1.2 and TLS 1.3

25. A browser displays a certificate warning. List five possible causes and explain how you would distinguish an expired
certificate from a hostname mismatch or an untrusted issuer.

Expired Certificate: The certificate's validity period has ended. Check the Not After date.

Hostname Mismatch: The requested domain does not match the certificate's Subject Alternative Name (SAN) entries.

Untrusted Issuer: The certificate is self-signed or issued by a Certificate Authority that the client does not trust. Check the issuer and certificate chain.

Certificate Not Yet Valid: The current date is earlier than the certificate's Not Before date. Also verify the system clock.

Incomplete Certificate Chain: The server may not provide the required intermediate certificates.

To distinguish these issues, inspect the certificate's validity dates, SAN entries, issuer, and certificate chain. Tools such as openssl s_client and curl -v can help investigate certificate errors

26. Explain why a valid HTTPS certificate does not automatically mean that a website is safe, bug-free, or trustworthy.

A valid HTTPS certificate does not automatically mean that a website is safe, bug-free, or trustworthy.
HTTPS and TLS help protect communication by providing encryption, server authentication, and data integrity. However, they do not guarantee the quality or honesty of the website itself.
For example:

A phishing website can still have a valid HTTPS certificate.

A website may contain malware or malicious scripts.

A website may have security vulnerabilities or software bugs.

The website owner may misuse users' personal information.
Therefore, users should also evaluate the website's reputation, content, security practices, and authenticity instead of relying only on the HTTPS padlock or certificate.

PART F – SSH & Secure Server Access

27. Inspect your SSH directory with ls -la ~/.ssh. Identify which files are public keys, private keys, and known-host
records if they exist. Do not upload or share private-key contents.

The ~/.ssh directory stores SSH keys and configuration files.

- **Public key files usually end with .pub.**
- **Private key files do not have the .pub extension.**

- **known_hosts stores information about previously connected SSH servers.**
- **authorized_keys contains public keys allowed to authenticate to a user account.**

28. If you already have a suitable key pair, explain how you would check its fingerprint using ssh-keygen -lf ~/.ssh/id_ed25519.pub. If no key exists, generate a new Ed25519 key pair using a filename of your choice and a passphrase.

The SSH key fingerprint is checked using ssh-keygen -lf. If no key pair exists, a new Ed25519 key pair can be generated using ssh-keygen -t ed25519. A passphrase adds an extra layer of security to the private key.
Command: ssh-keygen -lf ~/.ssh/id_ed25519.pub

29. Explain the roles of the SSH client, SSH server, public key, private key, and authorized_keys file during public-key authentication.

Component	Role
SSH Client	Initiates a secure connection to the server.
SSH Server	Receives SSH connection requests and authenticates clients.
Public Key	Shared with the server to verify the user's identity.
Private Key	Kept secret by the client and used to prove ownership of the public key.
authorized_keys	Stores public keys allowed to access a particular user account.

1. **How public-key authentication works:**
2. **SSH client sends a connection request.**
3. **SSH server checks the public key against authorized_keys.**

4. **Server sends an authentication challenge.**
5. **Client proves it owns the matching private key.**
6. **Server verifies the proof and grants access.**

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30. Write the SSH command to connect to a cloud VM using username ubuntu, host 203.0.113.25, and key file ./training-key.pem. Explain why the private key must be protected and how restrictive file permissions help.

ssh -i ./training-key.pem ubuntu@203.0.113.25

A private key must be protected because anyone who obtains it may be able to access the server as its owner.

Restrictive file permissions prevent unauthorized users from reading the private key.

chmod 400 training-key.pem

31. A connection fails with Permission denied (publickey). Give a troubleshooting checklist covering username, key selection, server-side authorized key, file permissions, and server configuration.

If an SSH connection fails with Permission denied (publickey), the following checklist can be used:

- **Username: Verify that the correct username is used, such as ubuntu.**
- **Key Selection: Ensure that the correct private key is specified using the -i option.**
- **authorized_keys: Check whether the matching public key is present in the server's ~/.ssh/authorized_keys file.**
- **File Permissions: Ensure that the private key has restrictive permissions.**

chmod 400 training-key.pem

32. Explain the difference between a network timeout, connection refused, and public-key authentication failure. What does each suggest about where the failure occurred?

Error	Meaning	Possible reason
Network timeout	**The server does not respond within the expected time.**	**Firewall, network issue, or server unreachable.**
Connection refused	**The server is reachable, but the connection is rejected.**	**SSH service is not running or port 22 is closed.**
Permission denied (publickey)	**The server is reachable, but authentication fails.**	**Wrong username, incorrect key, or missing public key.**

Explanation:

- **Network timeout: The failure usually occurs at the network connectivity level.**
- **Connection refused: The failure occurs at the server's connection or SSH service level.**
- **Permission denied (publickey): The connection reaches the SSH server, but authentication fails.**

PART G – Custom Domain & Website Delivery

Use an existing project or create a small static test page. Do not purchase a domain solely for this assignment; use a

domain you already control, a permitted free subdomain, or document the configuration without changing production DNS.

33. Draw a request-flow diagram for a visitor opening https://www.yourdomain.example. Include DNS lookup,

destination IP, HTTPS/TLS connection, hosting platform or reverse proxy, application response, and browser rendering.

When a visitor opens https://www.yourdomain.example, the browser first performs a DNS lookup to find the IP address associated with the domain.

fter receiving the destination IP address, the browser establishes an HTTPS connection with the server using the TLS handshake.

The hosting platform or reverse proxy receives the request and forwards it to the appropriate application.

The application processes the request and sends an HTTP response containing the website content.

Finally, the browser receives the response, processes the HTML, CSS, and JavaScript files, and renders the website for the visitor.

34. Prepare a sample DNS configuration table for a root domain and www subdomain. Include record type, host/name,

destination/value, and purpose. Clearly label it as an example if it is not your live configuration.

The following table shows an example DNS configuration for connecting a custom domain to a hosting platform. These are sample values and may differ from the actual DNS records provided by the hosting provider.

Record Type	Host/Name	Destination/Value	Purpose
A	@	76.76.21.21	Points the root domain to the hosting server.
CNAME	www	cname.vercel-dns.com	Points the www subdomain to the hosting provider.

The A record connects the root domain to an IPv4 address, while the CNAME record points the www subdomain to another hostname managed by the hosting provider. These records help visitors access the website using both the root domain and www subdomain

35. Explain why a root domain and www may require separate DNS records or provider-specific configuration. Describe

how you would verify both hostnames resolve and serve the intended site.

A root domain and a www subdomain may require separate DNS records because they are treated as different hostnames.

The root domain usually uses an A or ALIAS record to point to the hosting provider, while the www subdomain commonly uses a CNAME record.

Both hostnames must be configured correctly in the hosting platform to serve the same website.

DNS resolution can be verified using the nslookup yourdomain.com and nslookup www.yourdomain.com commands.

The website can also be tested by opening both hostnames in a browser and checking whether they display the intended website with a valid HTTPS connection.

36. A deployment is live on the hosting provider's default URL, but the custom domain shows an error. Create a

troubleshooting order covering DNS records, propagation/TTL, domain verification, HTTPS certificate provisioning,

redirects, and hosting configuration.

If a website works on the hosting provider's default URL but shows an error on the custom domain, the first step is to check whether the DNS records are correctly configured according to the hosting provider's instructions. Next, check DNS propagation and TTL to ensure that the updated records have reached DNS servers. Then verify that the custom domain has been successfully added and verified in the hosting platform. After that, check whether the HTTPS certificate has been provisioned correctly and whether the domain has a valid SSL/TLS configuration. Next, verify that redirects between the root domain and

www are configured correctly. Finally, check the hosting platform's domain settings to ensure that both hostnames are connected to the correct project and deployment.

37. Explain the difference between DNS resolution and an HTTP redirect. Give an example where DNS is correct but the

browser still lands on an unexpected URL.

DNS resolution is the process of translating a domain name into an IP address so that the browser can locate the server hosting the website. An HTTP redirect is a response sent by the web server that instructs the browser to navigate to a different URL. DNS resolution happens before the browser sends an HTTP request, whereas an HTTP redirect happens after the server receives the request.

For example, if www.example.com correctly resolves to the hosting server but the server is configured to redirect visitors to example.com, the browser will automatically open the root domain instead of the www subdomain. In this case, DNS is working correctly, but the HTTP redirect determines the final URL displayed in the browser.

PART H – Production Incident Challenges

38. Incident A – Website unreachable: Users report that a deployed website is down. Your machine can ping a public
IP, but curl -I https://yourdomain.example fails. Write a step-by-step investigation plan. Include the
commands you would run and what each result would prove or rule out.

If users report that a deployed website is down, I would investigate the issue step by step to identify whether it is related to DNS, network connectivity, HTTPS, or the hosting server.

First, I would use ping 8.8.8.8 to check basic network connectivity. If the ping is successful, it indicates that the machine can reach a public IP address, although it does not prove that the website is reachable.

Next, I would use nslookup yourdomain.example to check whether the domain resolves to an IP address.

I would use curl -I https://yourdomain.example to check the HTTP response and identify errors such as connection timeout or server errors.

Then, I would use curl -v https://yourdomain.example to inspect the connection and TLS handshake details.

I would also use traceroute yourdomain.example or tracepath yourdomain.example to identify possible network routing problems.

Finally, I would check the hosting provider's deployment status, server health, firewall rules, and web server logs to identify the root cause.

39. Incident B – DNS points to the wrong server: The hosting provider shows the correct deployment, but DNS lookup
returns an old IP. Explain how you would confirm the authoritative answer, inspect the configured records, and determine
whether caching or incorrect DNS configuration is responsible.

If the hosting provider shows the correct deployment but DNS returns an old IP address,
I would first use nslookup yourdomain.example and dig yourdomain.example to identify the IP address returned by DNS.

Then, I would use dig NS yourdomain.example to identify the authoritative nameservers responsible for the domain. After that, I would query an authoritative nameserver directly using dig @nameserver yourdomain.example A to verify the actual DNS record.

I would compare the returned IP address with the correct IP address provided by the hosting provider. If the authoritative answer contains the old IP, it indicates an incorrect DNS configuration.

If the authoritative answer is correct but other DNS resolvers return the old IP, it may be caused by DNS caching or propagation delays.

I would also inspect the DNS records in the domain registrar or DNS management dashboard and verify the record type, hostname, destination value, and TTL before making any changes.

40. Incident C – SSH works from one network only: An engineer can SSH to a VM from the office but not from home.
Identify likely network, firewall, routing, and access-control causes. Explain what evidence you would collect before
changing rules.

If an engineer can connect to a virtual machine using SSH from the office but cannot connect from home, the issue may be related to network restrictions, firewall rules, routing, or access control.

First, I would check whether the home network has internet connectivity using ping 8.8.8.8. Next, I would test SSH port connectivity using nc -vz <server-ip> 22. If the connection times out, it may indicate that a firewall or network rule is blocking port 22.

I would inspect the cloud security group, network firewall, and server firewall to check whether SSH access is restricted to the office public IP address. I would also check whether the home network or internet service provider blocks outbound SSH traffic.

Additionally, I would verify routing, VPN requirements, and SSH access-control settings. Before changing any firewall rules, I would collect evidence such as connection error messages, source public IP addresses, firewall logs, security group rules, and SSH server logs to identify the actual cause.

41. Incident D – HTTPS failure after deployment: A custom domain loads over HTTP but HTTPS shows a certificate
error. Describe the checks you would perform and how you would verify the final fix.

If a custom domain loads over HTTP but HTTPS shows a certificate error, I would first verify that the domain is correctly connected to the hosting provider.

Next, I would check whether the SSL/TLS certificate has been successfully issued and whether it covers both the root domain and the www subdomain. I would use nslookup yourdomain.example to verify DNS resolution and ensure that the domain points to the correct hosting server. Then, I would use curl -Iv https://yourdomain.example

to inspect the HTTPS connection and identify certificate-related errors. I would also use openssl s_client -connect yourdomain.example:443 -servername yourdomain.example to inspect the certificate details, validity period, and hostname.

If the certificate is missing, expired, or does not match the domain, I would check the hosting provider's SSL configuration and DNS verification status. After correcting the configuration, I would verify the fix by opening the website using HTTPS in a browser and checking that the connection is secure and the certificate is valid

42. Incident E – Application server is listening, but users cannot connect: The service listens on port 3000 on a VM.
Explain how you would investigate the application bind address, host firewall, cloud security rules, routing, and
reverse-proxy configuration.

- **If an application server is listening on port 3000 but users cannot connect, I would investigate the application, firewall, cloud security rules, network routing, and reverse-proxy configuration.**
- **First, I would run ss -tulnp | grep :3000 to check whether the application is listening on port 3000. Then, I would use curl http://localhost:3000 to verify whether the application responds locally.**
- **I would inspect the application bind address to ensure that it listens on the correct network interface.**
- **If it listens only on 127.0.0.1, external users cannot access it directly. The application may need to listen on 0.0.0.0 when direct external access is intended. Next, I would check the host firewall using sudo ufw status and inspect cloud security group rules to verify whether the required port is allowed.**
- **I would also check network routing and ensure that traffic can reach the VM. Finally, if Nginx or Apache is used as a reverse proxy, I would verify its configuration, upstream address, port number, and service status. I would test the application again using curl and check server logs to confirm that users can successfully access the application.**